Correcting Length Bias: Replicating and Stress-Testing Pivoted Document Length Normalization

Information Retrieval, Intermediate Status Update
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Code: https://github.com/musaddi-drishya/Info_Ret_Project

1. Implementation Status and Architecture

We have built the complete retrieval system from scratch in Python, using only NumPy, Pandas and SQLite (no scikit-learn, Lucene or Elasticsearch for the core algorithm). All five planned experiments (E1 to E5) are running on both public datasets, and we also finished the two stretch experiments (pivoted unique normalization and BM25) earlier than planned. A full run of every experiment takes about 15 seconds. The architecture is the same as in our proposal, with two additions: a judging module for the campus dataset, and the two extra scoring functions, which reuse the same index.

Component	Status	Notes
Preprocessing: tokenizer, stopwords, Porter stemmer	Done	Stemmer written ourselves and checked against known outputs
SQLite storage, inverted index, .npz cache	Done	Postings stored as NumPy arrays for fast scoring
TF-IDF with cosine (E1) and pivoted cosine (E2)	Done	Pivot set to the average cosine norm
20 Newsgroups and Reuters samples, 55 queries	Done	1,000 documents each, balanced by topic and length
Evaluation metrics, length-bias analysis, timing	Done	P, R and F1 at 5, 10 and 20; MAP; NDCG@10
Verbosity (E3) and multi-topic (E4) attacks	Done	Synthetic documents with known relevance
Slope sweep (E5), with and without attack	Done	Slope from 0 to 1 in steps of 0.1
Stretch: pivoted unique (E6), BM25 (E7)	Done	Planned for week 6, finished early
Campus dataset	Collected	100 BITS Pilani news and notice pages; 205 pairs ready for judging
Failure analysis, OCR-noise stretch, demo	Pending	See Section 4

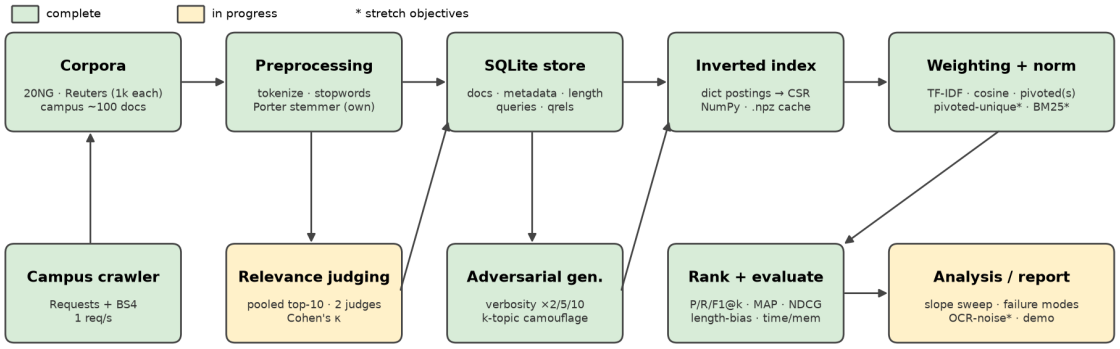


Figure 1: System architecture. Green parts are complete, yellow parts are in progress.

2. IR Pipeline

2.1 Data

For 20 Newsgroups we removed the message headers, since they contain the newsgroup name and would give away the answer, but kept the subject line. We also removed quoted replies and signatures. From Reuters-21578 we used only documents with a single topic, from 20 topics. We sampled 1,000 documents from each dataset so that every category and every length range is represented, and dropped very short documents (under 20 terms). We wrote 30 queries for 20 Newsgroups and 25 for Reuters; as stated in our proposal, a document counts as relevant if it belongs to the query's category.

The campus dataset was collected with a small crawler (Requests and BeautifulSoup, one request per second) that removes duplicate notices. The campus portal does not post 100 items in a single week, so our 100 pages cover January 2025 to October 2026. We have written 18 queries for it and prepared 205 query-document pairs for two of us to judge.

2.2 Preprocessing and indexing

Text is lower-cased and split into words, stopwords are removed, and words are stemmed with our own implementation of the Porter stemmer. The inverted index is first built as a dictionary of posting lists and then converted to NumPy arrays, so that scoring a query term is a single array operation. For each document we also store its length, number of unique terms and average term frequency. The 20 Newsgroups index has 19,257 terms and

81,003 postings, takes 2.1 MB and is built in 1.8 s. The Reuters index has 6,984 terms, takes 1.0 MB and is built in 0.8 s.

2.3 Retrieval models

All models score a document as the sum, over query terms, of the query weight times the document weight, divided by a normalization factor N(d). Term weights use w = (1 + ln tf) × ln(N/df). The models differ only in N(d):

$$\text{Cosine: } N(d) = \|d\| \qquad \text{Pivoted cosine: } N(d) = (1 - s) \times p + s \times \|d\|$$

Here p, the pivot, is the average value of ||d|| over the collection, and s is the slope (we use s = 0.7 by default; s = 1 gives plain cosine). For pivoted unique normalization we follow Singhal et al. and use the number of unique terms in the document instead of ||d||, with s = 0.2. For BM25 we use k1 = 1.2 and b = 0.75.

2.4 Evaluation and attacks

We report P@k, R@k and F1@k for k = 5, 10 and 20, MAP, NDCG@10, query time and index size. To measure length bias we use the method from the original paper: documents are sorted into 10 bins by length, and for each bin we compare how often it is retrieved (top 50 per query) with how often it is relevant. We summarize the difference with two numbers: the total variation distance (TVD) between the two distributions, and the log-length gap, which is positive when retrieved documents are longer than relevant ones. For the verbosity attack (E3) we replace 50 random documents with copies of themselves repeated 2, 5 or 10 times. For the multi-topic attack (E4) we add 20 synthetic documents of 300 terms each, made from 1, 2, 3 or 4 topics, plus a control made by repeating one short document.

3. Preliminary Results

Dataset	Model	P@10	R@10	F1@10	P@20	MAP	NDCG@10	TVD	Len. gap	ms/query
20NG	Cosine (baseline)	0.713	0.143	0.238	0.657	0.373	0.742	0.093	+0.171	0.08
20NG	Pivoted, s = 0.7	0.730	0.146	0.243	0.655	0.374	0.757	0.144	+0.292	0.06
20NG	Pivoted unique	0.740	0.148	0.247	0.652	0.371	0.756	0.171	+0.336	0.08
20NG	BM25	0.733	0.147	0.244	0.645	0.374	0.754	0.139	+0.277	0.09
Reuters	Cosine (baseline)	0.932	0.184	0.307	0.858	0.714	0.946	0.050	+0.052	0.20
Reuters	Pivoted, s = 0.7	0.920	0.182	0.304	0.850	0.714	0.935	0.089	+0.133	0.15
Reuters	Pivoted unique	0.912	0.180	0.301	0.840	0.708	0.933	0.127	+0.197	0.15
Reuters	BM25	0.952	0.188	0.314	0.882	0.725	0.965	0.106	+0.152	0.11

Table 1: Results on the original (unattacked) datasets. Pivoted unique and BM25 are our stretch models.

3.1 Length bias (RQ1 and RQ2)

We expected cosine normalization to favor short documents, as the original paper found on TREC. This is not what we see. On both datasets cosine already retrieves long documents slightly more often than they are relevant (log-length gap of +0.171 on 20 Newsgroups and +0.052 on Reuters, Figure 2 left). Pivoting pushes retrieval further towards long documents (TVD goes from 0.093 to 0.144 on 20 Newsgroups), while MAP hardly changes (0.3729 against 0.3735). The best slopes were 0.7 and 0.8. We think the reason is our setup: because we sampled evenly across lengths and use categories as relevance, long documents are not more likely to be relevant here, whereas pivoting assumes they are. This is one of the main reasons we want results on the human-judged campus dataset.

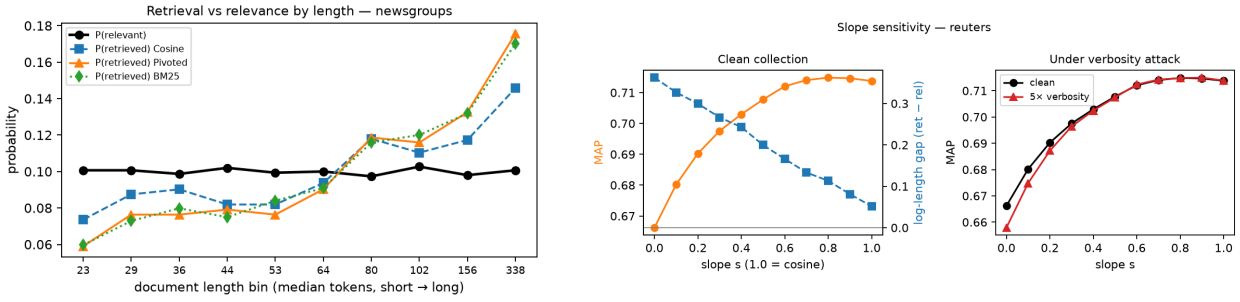


Figure 2: Left, retrieval against relevance by length bin on 20 Newsgroups. Right, MAP and log-length gap against the slope on Reuters, with and without 5x verbosity padding.

3.2 Verbosity attack (RQ3)

Model	P@10	R@10	F1@10	MAP	Median rank of padded docs	Wrong docs in top 10
Cosine	0.932 / 0.924	0.184 / 0.183	0.307 / 0.305	0.7138 / 0.7136	34 / 38	0 / 0
Pivoted, s = 0.7	0.920 / 0.920	0.182 / 0.182	0.304 / 0.304	0.7140 / 0.7140	36 / 30	1 / 2
Pivoted unique	0.912 / 0.912	0.180 / 0.180	0.301 / 0.301	0.7077 / 0.7068	45 / 45	3 / 2
BM25	0.952 / 0.944	0.188 / 0.187	0.314 / 0.312	0.7246 / 0.7178	36 / 28	2 / 4

Table 2: Verbosity attack on Reuters, before / after padding 50 documents to 10 times their length. The last column counts padded documents that appear in the top 10 of queries they are not relevant to.

Repeating a document adds no new information, so ideally its rank should not change. Cosine slightly lowers padded documents, while pivoted cosine and BM25 raise them. Pivoted unique normalization barely reacts, because repeating text does not change the number of unique terms. Precision, recall and MAP change by less than 0.01, since only 5% of the documents were attacked, but the effect gets much stronger as the slope decreases (Figure 2 right). At s = 0 the padded documents move from median rank 44 to 14 and appear 14 times in the top 10 of unrelated queries, compared with 1 at s = 0.7 and 0 at s = 1.

3.3 Multi-topic documents (RQ4) and slope (RQ5)

Injected document (300 terms)	Cosine	Pivoted, s = 0.7	Pivoted, s = 0.3	Pivoted unique	BM25
One short document repeated	31	22	14	38	21
1 topic	19	14	7	11	12
2 topics	36	31	18	25	26
3 topics	41	35	27	31	34
4 topics	62	54	41	50	51

Table 3: Median rank of the injected documents on Reuters, for the queries they are relevant to (lower is better).

When all documents have the same length, every model ranks documents covering several topics much lower than single-topic ones (with cosine, the median rank goes from 19 for one topic to 62 for four). Each topic only gets a quarter of the document, and length normalization cannot tell this apart from padding. A lower slope helps the multi-topic documents, but it helps the repeated control document in the same way (at s = 0.3 it reaches rank 14, against 18 for two-topic documents). For the slope itself, MAP is almost flat between 0.6 and 1.0 and drops quickly below 0.5, which is also where the padding attack starts to work. So the slope that works best on clean data is also the one that limits the attack.

4. Plan for the Final Submission

We estimate that about 20 to 25% of the work is left. The table below shows what remains and who is responsible.

Task	Owner	Target date
Campus dataset: finish the queries, judge the 205 pairs independently, compute Cohen's kappa, and run E1 to E5 on it	Drishya, Aditya	1 to 3 Nov
Choose the slope on separate queries (so the result is not tuned on the test set) and add significance tests for Table 1	Divyansh	3 to 5 Nov
Failure analysis: look at individual queries where attacks succeed; test pivoted unique on multi-topic documents	Himabh	3 to 7 Nov
Stretch: OCR-like noise and its effect on length normalization	Himabh, Aditya	7 to 10 Nov
Final report, slides and demo	All	10 to 14 Nov

Stretch objectives. Pivoted unique normalization and BM25 (including a sweep over its b parameter) are finished and included in the results above. The OCR-noise experiment has not been started yet.

5. Contribution Matrix

Member	Components owned and completed so far	Status
Divyansh Jain	Inverted index and .npz caching; the four scoring models (TF-IDF cosine, pivoted cosine, pivoted unique, BM25); experiment runner and search tool	Done
Drishya Musaddi	Loading and cleaning 20 Newsgroups and Reuters, stratified sampling, queries; SQLite storage; campus crawler and the judging pool	Done, campus judging in progress
Aditya Goyal	Preprocessing and our own Porter stemmer; evaluation metrics; length-bias analysis; plots and tests	Done
Himabh Gemini	Adversarial document generation (verbosity padding, multi-topic documents); attack experiments; attack demo; this report	Done, failure analysis ongoing

References

[1] A. Singhal, C. Buckley and M. Mitra. Pivoted Document Length Normalization. In Proceedings of SIGIR, 1996.

[2] S.-H. Na, I.-S. Kang and J.-H. Lee. Improving Term Frequency Normalization for Multi-topical Documents and Application to Language Modeling Approaches. arXiv:1502.02277.

[3] S. E. Robertson and S. Walker. Some Simple Effective Approximations to the 2-Poisson Model for Probabilistic Weighted Retrieval. In Proceedings of SIGIR, 1994.